METRICS AND APPROXIMABILITY FOR TRIANGULATED CATEGORIES

Introduction

Triangulated categories are ubiquitous in algebra, topology, and geometry. Neeman has recently introduced certain metric/analytic techniques for triangulated categories in a series of articles [Nee21c, Nee18, Nee26, BNP23, Nee24a, CHNS24, Nee25][1], and used them to settle multiple conjectures concerning triangulated categories arising in algebraic geometry, and to prove other new results[2].

The idea of this mini-course is to give a feeling of these techniques in action, by going through some of the interesting results proven using them. Most of the applications so far are in algebraic geometry, although these techniques are slowly being applied in representation theory and homotopy theory also, see for example [CG24, Mat26a, Mat26b, LS25].

In particular, we will look at the connection between the regularity of a Noetherian scheme with the existence of bounded t-structures and strong generators for the category of perfect complexes. Further, we will discuss some new representability theorems and how to use them to give a unified proof of GAGA theorems in algebraic geometry.

In the second half, we will prove that the derived category $D_{\mathrm{Qcoh}}(X)$ of a Noetherian separated scheme is approximable. We will then move on to the passage between various triangulated subcategories of a weakly approximable triangulated category, which is a vast generalisation of a result by Rickard on derived Morita theory. Finally, we will end with a result on the bijection of semiorthogonal decompositions on various (small) triangulated categories associated to a scheme.

Prerequisites. We will assume familiarity with the basics of triangulated categories, or their enhanced incarnations (stable model/$(\infty,1)$-categories, pretriangulated DG-categories $\dots$). As most of the applications we will discuss are algebro-geometric, we will expect some familiarity with the language of schemes, and with the derived categories and functors associated to them.

Conventions. We will use the following conventions and notations without mention in the rest of the document.

(1) All rings will be commutative unless explicitly mentioned otherwise.
(2) Our gradings will be cohomological. For full subcategories $\mathcal{A}_1, \mathcal{A}_2$ of a triangulated category $\mathcal{T}$, we denote by $\mathcal{A}_1 * \mathcal{A}_2$ to be the full subcategory consisting of objects A for which there exists a triangle $A_1 \to A \to A_2 \to A_1[1]$ with $A_i \in \mathcal{A}_i$.
(3) When working with triangulated categories associated to a scheme, we will reserve the notation $\tau^{\leq 0}, \tau^{\geq 0}$ and $\sigma^{\leq 0}, \sigma^{\geq 0}$ for the so called canonical and stupid truncations respectively [Sta24, Tag 0118].
(4) We will use "exact triangles" and "triangles" interchangeably.

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[1]In the order the preprints appeared on arXiv.

[2]See also the survey articles [Nee20, Nee21a, Nee23, CNS25].

1. Bounded t-structures and regularity

Originally introduced in the early 80's by Beilinson, Bernstein, and Deligne to study perverse sheaves, t-structures have by now become important tools to study triangulated categories. Of particular importance are bounded t-structures, which play a central role in the theory of Bridgeland stability conditions. Further, in the theory of motives, the conjectured motivic (bounded) t-structure is of central importance. For completeness, we begin by the definition, followed by a prototypical example.

Definition 1.1. Let $\mathcal{T}$ be a triangulated category. A *t-structure* on $\mathcal{T}$ is a pair of strictly full subcategories $(\mathcal{T}^{\leq 0}, \mathcal{T}^{\geq 0})$[3] such that

(1) $\mathcal{T}^{\leq 0} \subseteq \mathcal{T}^{\leq 0}[-1]$ and $\mathcal{T}^{\geq 0} \subseteq \mathcal{T}^{\geq 0}[1]$,
(2) $\operatorname{Hom}(A, B) = 0$ for all $A \in \mathcal{T}^{\leq 0}$ and $B \in \mathcal{T}^{\geq 0}[-1]$,
(3) for all $T \in \mathcal{T}$, there exists a triangle $A \to T \to B \to A[1]$ with $A \in \mathcal{T}^{\leq 0}$ and $B \in \mathcal{T}^{\geq 0}[-1]$.

For notational convenience, we define $\mathcal{T}^{\leq n} := \mathcal{T}^{\leq 0}[-n]$ and $\mathcal{T}^{\geq n} := \mathcal{T}^{\geq 0}[-n]$. Finally, we say that a t-structure is *bounded* if for all $T \in \mathcal{T}$, there exists $n(T) \geq 0$ with $T \in \mathcal{T}^{\geq -n(T)} \cap \mathcal{T}^{\leq n(T)}$.

Remark 1.2. It turns out that the triangles we get in Definition 1.1 are functorial. In fact, we get a functor $\tau^{\leq 0} \colon \mathcal{T} \to \mathcal{T}^{\leq 0}$ (resp. $\tau^{\geq 0} \colon \mathcal{T} \to \mathcal{T}^{\geq 0}$) which is the right (resp. left) adjoint to the inclusion. Furthermore, we get a triangle $\tau^{\leq 0}T \to T \to \tau^{\geq 0}T \to \tau^{\leq 0}T[1]$ for any $T \in \mathcal{T}$ with the first (resp. second) map given by the counit (resp. unit) of adjunction. This triangle, called the *truncation triangle for T*, is canonically isomorphic to the triangle in Definition 1.1(3).

It turns out that the intersection $\mathcal{T}^{\heartsuit} := \mathcal{T}^{\leq 0} \cap \mathcal{T}^{\geq 0}$, called the *heart of the t-structure*, is an abelian category. Furthermore, the functor

$$\tau^{\leq 0} \circ \tau^{\geq 0} \simeq \tau^{\geq 0} \circ \tau^{\leq 0} \colon \mathcal{T} \to \mathcal{T}^{\heartsuit}$$

is a homological functor, that is, it sends triangles to long exact sequences.

Example 1.3. Let $\mathcal{A}$ be an abelian category and $\mathcal{B} \subseteq \mathcal{A}$ any Serre subcategory[4]. Then, $D_{\mathcal{B}}(\mathcal{A})$, which is the derived category of $\mathcal{A}$ with cohomology in $\mathcal{B}$, has a t-structure[5] $(D_{\mathcal{B}}(\mathcal{A})^{\leq 0}, D_{\mathcal{B}}(\mathcal{A})^{\geq 0})$ defined by

$$D_{\mathcal{B}}(\mathcal{A})^{\leq 0} = \{F : H^i(F) = 0 \text{ for all } i \geq 1\},\ D_{\mathcal{B}}(\mathcal{A})^{\geq 0} = \{F : H^i(F) = 0 \text{ for all } i \leq -1\},$$

with truncation triangles given by the so called canonical truncation [Sta24, Tag 0118]. This restricts to a bounded t-structure on $D^b_{\mathcal{B}}(\mathcal{A}) = \{F : H^i(F) = 0 \text{ for } |i| \gg 0\}$.

In particular, let X be a Noetherian scheme. Let $\mathcal{A} = \operatorname{Mod}(\mathcal{O}_X)$ be the (Grothendieck) abelian category of $\mathcal{O}_X$-modules. Then,

(1) taking $\mathcal{B} = \operatorname{Qcoh}(X)$ the Serre subcategory of quasi-coherent sheaves, we get a t-structure $(D_{\mathrm{Qcoh}}(X)^{\leq 0}, D_{\mathrm{Qcoh}}(X)^{\geq 0})$ defined by

$$D_{\mathrm{Qcoh}}(X)^{\leq 0} = \{F : \mathcal{H}^i(F) = 0 \text{ for all } i \geq 1\},\ D_{\mathrm{Qcoh}}(X)^{\geq 0} = \{F : \mathcal{H}^i(F) = 0 \text{ for all } i \leq -1\}.$$

(2) taking $\mathcal{B} = \operatorname{coh}(X)$ the Serre subcategory of coherent sheaves, this t-structure restricts to a bounded t-structure on $D^b_{\mathrm{coh}}(X)$, the derived category of sheaves with bounded and coherent cohomology.

[3]Although sometimes we will denote a t-structure by $(\mathcal{U}, \mathcal{V})$.
[4]That is, an abelian subcategory such that for any exact $M_1 \to M \to M_2$, $M_1, M_2 \in \mathcal{B} \implies M \in \mathcal{B}$.
[5]This is well known, see for example [GM03, IV.4, Proposition 3].

We begin our story[6] with a beautiful result by Antieau, Gepner, and Heller which provides an obstruction to the existence of bounded t-structures.

Theorem 1.4 ([AGH19, Theorem 1.1])**.** *If $\mathcal{T}$ is an (enhanced) essentially small triangulated category with a bounded t-structure then the first negative K-theory of $\mathcal{T}$ vanishes.*

Based on this result, the authors proposed the following conjecture.

Conjecture 1.5 ([AGH19, Conjecture 1.5])**.** Let X be a singular Noetherian finite-dimensional scheme. Then, $\mathrm{Perf}(X)$ has no bounded t-structures.

Let us review some of the evidence for (and against) the conjecture from what is known about the K-theory of schemes[7].

Discussion 1.6. Let X be a Noetherian scheme. Then,

(1) if X is regular then $K_{-n}(X) = 0$ for all $n \geq 1$ by the work of Schlichting [Sch06, Example 9.6],
(2) if X is singular, the negative K-groups are often non-trivial,
(3) but, if X is 0-dimensional, then $K_{-n}(X) = 0$ for $n \geq 1$.

The above might not instill too much belief for the conjecture, but thanks to Neeman, we now know that it is in fact true! Moreover, Neeman proves a relative version; to state it recall that for a closed subset Z of a Noetherian scheme X, $\mathrm{Perf}_Z(X)$ denotes the category of perfect complexes supported on Z.

Theorem 1.7 ([Nee24a, Theorem 0.1])**.** *Let X be a Noetherian finite-dimensional scheme and $Z \subseteq X$ a closed subset. Then, $\mathrm{Perf}_Z(X)$ has a bounded t-structure if and only if $Z \subseteq \mathrm{reg}(X)$.*

The proof of this result will occupy the rest of this section. Let us begin with the following categorical characterisation of regularity[8].

Lemma 1.8. *Let X be a Noetherian scheme and Z a closed subscheme. Then, $Z \subseteq \mathrm{reg}(X)$ if and only if the natural inclusion $\mathrm{Perf}_Z(X) \to D^b_{\mathrm{coh},Z}(X)$ is an equivalence.*

Proof. Assume that $Z \subseteq \mathrm{reg}(X)$. Let $F \in D^b_{\mathrm{coh},Z}(X)$. By our assumption, we know that either F_x is zero or a bounded complex with finitely generated cohomology over a regular local ring and hence perfect[9]. As F is bounded, this implies that F is in fact a perfect complex[10].

Conversely, assume that the natural inclusion $\mathrm{Perf}_Z(X) \to D^b_{\mathrm{coh},Z}(X)$ is an equivalence. Let $x \in Z$ be any closed point. Consider the skyscraper sheaf $k(x)$ at x, which lies in $D^b_{\mathrm{coh},Z}(X)$. Hence, by our hypothesis, it lies in $\mathrm{Perf}_Z(X)$, which in turn implies that $k(x)$ has finite projective dimension as an $\mathcal{O}_{X,x}$-module. And so, $\mathrm{gl.dim.}(\mathcal{O}_{X,x}) = \mathrm{proj.dim.}(k(x)) < \infty$; that is, $\mathcal{O}_{X,x}$ is a regular local ring. As regularity is closed under generalisation, we get that $Z \subseteq \mathrm{reg}(X)$. □

[6]Firmly outside the author's comfort zone!

[7]For a detailed discussion, see the relevant portions of [Nee24b].

[8]Which is well-known to those who know it well.

[9]As can be seen by taking a projective resolution P of F_x by finitely generated projectives and considering $\sigma^{\leq n}P$ where n is the minimal non-zero cohomology of P.

[10]One way to show this is go affine local, and use an argument as the footnote above using the fact that for any module M and $n \geq 0$, the subset $\{x : \mathrm{proj.dim.}(M_x) \leq n\}$ is open, see [IT19, Lemma 2.3].

Remark 1.9. In light of the previous result, we need to show that $\mathrm{Perf}_Z(X)$ has a bounded t-structure if and only if the natural inclusion $\mathrm{Perf}_Z(X) \to D^b_{\mathrm{coh},Z}(X)$ is an equivalence. As $D^b_{\mathrm{coh},Z}(X)$ always has a bounded t-structure, one of the implications is trivially satisfied. What remains to show is that $\mathrm{Perf}_Z(X)$ having a bounded t-structure implies that the natural inclusion $\mathrm{Perf}_Z(X) \to D^b_{\mathrm{coh},Z}(X)$ is an equivalence.

A tale of two t-structures.

Notation 1.10. Suppose $\mathcal{A}$ is a subset of $\mathrm{Perf}_Z(X)$, and let $\mathcal{B} = \bigcup_{n\geq 0} \mathcal{A}[n]$. Then, we can define a t-structure $(\mathcal{T}^{\leq 0}_{\mathcal{A}}, \mathcal{T}^{\geq 0}_{\mathcal{A}})$ with $\mathcal{T}^{\leq 0}_{\mathcal{A}} := \mathrm{Coprod}(\mathcal{B})$ and $\mathcal{T}^{\geq 0}_{\mathcal{A}} := \mathcal{B}^{\perp}[1]$ on $D_{\mathrm{Qcoh},Z}(X)$ by [ATJLS03, Theorem A.1]. Here $\mathrm{Coprod}(\mathcal{B})$ denotes the subcategory of $\mathcal{T}$ generated by $\mathcal{B}$ which is closed under coproducts and extensions. Further, for any object $E \in \mathcal{T}$, we define $(\mathcal{T}^{\leq 0}_{E}, \mathcal{T}^{\geq 0}_{E}) := (\mathcal{T}^{\leq 0}_{\{E\}}, \mathcal{T}^{\geq 0}_{\{E\}})$.

Suppose $\mathrm{Perf}_Z(X)$ has a bounded t-structure $(\mathcal{U}, \mathcal{V})$. Then in the notation above, we can construct a t-structure $(\mathcal{T}^{\leq 0}_{\mathcal{U}}, \mathcal{T}^{\geq 0}_{\mathcal{U}})$ on $D_{\mathrm{Qcoh},Z}(X)$ which restricts back to the t-structure $(\mathcal{U}, \mathcal{V})$ on $\mathrm{Perf}_Z(X)$. Note that $D_{\mathrm{Qcoh},Z}(X)$ already has the so called "standard t-structure" defined via canonical truncations which we denote by $(D_{\mathrm{Qcoh},Z}(X)^{\leq 0}, D_{\mathrm{Qcoh},Z}(X)^{\geq 0})$. The main technical ingredient in proving our main result will be a comparison between these two t-structures. To show why this is enough, we begin by considering the toy case in which the two t-structures agree, or equivalently, when the standard t-structure restricts to the category of perfect complexes[11].

Proposition 1.11. *Let X be a Noetherian scheme and $Z \subseteq X$ a closed subset such that the standard t-structure on $D_{\mathrm{Qcoh},Z}(X)$ restricts to $\mathrm{Perf}_Z(X)$. Then, $Z \subseteq \mathrm{reg}(X)$, or equivalently, the natural inclusion $\mathrm{Perf}_Z(X) \to D^b_{\mathrm{coh},Z}(X)$ is an equivalence.*

We will look at two proofs of this. The first is perhaps a bit easier, but the second more amenable to a generalisation which we will need later.

Proof 1 of Proposition 1.11. Let $x \in Z$ be a closed point, and $j\colon U = \mathrm{Spec}(R) \to X$ be any affine open neighbourhood of x. Let $K(x)$ be the Koszul complex on x[12], and consider $\widetilde{K(x)} := Rj_*K(x)$. By staring intently[13] at the equivalence $Lj^*\colon D_{\mathrm{Qcoh},\{x\}}(X) \leftrightarrows D_{\mathrm{Qcoh},\{x\}}(U) : Rj_*$[14], we see that $\widetilde{K(x)} \in \mathrm{Perf}_Z(X)$ and $\tau^{\geq 0}\widetilde{K(x)} = k(x)$ where $k(x)$ denotes the skyscraper sheaf at x. So, by our hypothesis $k(x)$ is a perfect complex, and hence has finite projective dimension as a $\mathcal{O}_{X,x}$-module. And so, $\mathrm{gl.dim.}(\mathcal{O}_{X,x}) = \mathrm{proj.dim.}(k(x)) < \infty$; that is, $\mathcal{O}_{X,x}$ is a regular local ring. As regularity is closed under generalisation, we get that $Z \subseteq \mathrm{reg}(X)$. □

Before we get to the second proof, we need the following result which was originally proven for $Z = X$ in [LN07].

Lemma 1.12 ([Sta24, Tag 08EL])**.** *Let X be a Noetherian scheme with a closed subset Z. Then, for any $F \in D^-_{\mathrm{coh},Z}(X)$ and any integer n, there exists $E \in \mathrm{Perf}_Z(X)$ and a triangle $E \to F \to D \to E[1]$ with $D \in D_{\mathrm{Qcoh}}(X)^{\leq -n}$.*

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[11] See [BL26, Proposition 2.5] for a related result for ring spectra.

[12] That is, choose some finite set of generators $\{f_1, \cdots, f_n\}$ for the ideal defined by x, and set $K(x) := K(f_1, \cdots .f_r) = \bigotimes_{i=1}^r \mathrm{Cone}(R \xrightarrow{f_i} R)$.

[13] Petition to change the "it is easy to see" to more creative alternatives.

[14] See [Nee24a, Corollary 6.4] for a proof of this equivalence, though intuitively this should be clear as $\{x\} \subseteq U$.

This is in general a difficult and somewhat technical result. We give a proof for the special case $Z = X$ when there are enough vector bundles as an illustration.

Proof of Theorem 1.12 for $Z = X$ when there are enough vector bundles. By our hypothesis, for every coherent sheaf there is a vector bundle surjecting onto it. This almost immediately implies that for any object in $D^-_{\mathrm{coh}}(X)$ is quasi-isomorphic to a complex P such that $P^i = 0$ for $i > m$ and each P^i is a vector bundle. We then get the required triangles by the so called "brutal" or "hard" truncations of P. □

Proof 2 of Proposition 1.11. It is enough to show that every object in $D^b_{\mathrm{coh},Z}(X)$ is a perfect complex. Let $F \in D^b_{\mathrm{coh},Z}(X)$. By replacing F by $F[n]$ for some $n \ll 0$ if necessary, we may assume that $F \in D^b_{\mathrm{coh},Z}(X)^{\geq 0}$. By Theorem 1.12, there exists a triangle $E \to F \to D \to E[1]$ with $E \in \mathrm{Perf}_Z(X)$ and $D \in D_{\mathrm{Qcoh}}(X)^{\leq -2}$. Then, in the triangle $D[-1] \to E \to F \to D$ we have that $D[-1] \in D_{\mathrm{Qcoh}}(X)^{\leq -1}$ and $F \in D_{\mathrm{Qcoh}}(X)^{\geq 0}$. So, this must be the truncation triangle for E with respect to the standard t-structure[15]. Therefore, $F = \tau^{\geq 0}E$ is a perfect complex by our hypothesis. □

Now we move on to the more general case, where the two t-structures are not necessarily the same. We begin with the following definition which gives an equivalence relation on the collection of t-structures. We will see that the proof above works even if the two t-structures are merely equivalent. Then, the remaining task would be to show that the two t-structures are equivalent, regardless of the choice of the bounded t-structure on $\mathrm{Perf}_Z(X)$ we begin with.

Definition 1.13 ([Nee24a]). Two t-structures $(\mathcal{U}_1, \mathcal{V}_1)$ and $(\mathcal{U}_2, \mathcal{V}_2)$ on a triangulated category $\mathcal{T}$ are *equivalent* if there exists an integer $n \geq 0$ such that $\mathcal{U}_2[n] \subseteq \mathcal{U}_1 \subseteq \mathcal{U}_2[-n]$ (equivalently $\mathcal{V}_2[n] \supseteq \mathcal{V}_1 \supseteq \mathcal{V}_2[-n]$).

We now show that the proof above can be easily modified to the case when the two t-strucutures are equivalent.

Theorem 1.14. *Let X be a Noetherian scheme with a closed subset Z and $(\mathcal{U}, \mathcal{V})$ be a t-structure on $\mathrm{Perf}_Z(X)$ such that the t-structure $(\mathcal{T}^{\leq 0}_{\mathcal{U}}, \mathcal{T}^{\geq 0}_{\mathcal{U}})$ is equivalent to the standard t-structure on $D_{\mathrm{Qcoh},Z}(X)$. Then, the natural inclusion $\mathrm{Perf}_Z(X) \to D^b_{\mathrm{coh},Z}(X)$ is an equivalence.*

Proof. First, note that by the definition of equivalence of t-structures, there exists $N \geq 0$ with

$$D_{\mathrm{Qcoh},Z}(X)^{\leq -N} \subseteq \mathcal{T}^{\leq 0}_{\mathcal{U}} \subseteq D_{\mathrm{Qcoh},Z}(X)^{\leq N}.$$

It is enough to show that every object in $D^b_{\mathrm{coh},Z}(X)$ is a perfect complex. Let $F \in D^b_{\mathrm{coh},Z}(X)$. By replacing F by $F[n]$ for some $n \ll 0$ if necessary, we may assume that $F \in D^b_{\mathrm{coh},Z}(X)^{\geq N} \subseteq \mathcal{T}^{\geq 0}_{\mathcal{U}}$. By Theorem 1.12, there exists a triangle $E \to F \to D \to E[1]$ with $E \in \mathrm{Perf}_Z(X)$ and $D \in D_{\mathrm{Qcoh}}(X)^{\leq -N-2}$. Then, in the triangle $D[-1] \to E \to F \to D$ we have that $D[-1] \in D_{\mathrm{Qcoh}}(X)^{\leq -N-1} \subseteq \mathcal{T}^{\leq -1}_{\mathcal{U}}$ and $F \in D_{\mathrm{Qcoh}}(X)^{\geq 0}$. So, this must be the truncation triangle for E with respect to the t-structure $(\mathcal{T}^{\leq 0}_{\mathcal{U}}, \mathcal{T}^{\geq 0}_{\mathcal{U}})$. By the uniqueness of truncation triangles, F is a perfect complex. □

[15]Recall that for any t-structure $(\mathcal{T}^{\leq 0}, \mathcal{T}^{\geq 0})$ and for any $T \in \mathcal{T}$, there is a *unique* triangle $A \to T \to B \to A[1]$ with $A \in \mathcal{T}^{\leq 0}$ and $B \in \mathcal{T}^{\geq 0}$.

So, all that remains to be shown is that for any bounded t-structure $(\mathcal{U}, \mathcal{V})$ on $\mathrm{Perf}_Z(X)$, the induced t-structure $(\mathcal{T}_{\mathcal{U}}^{\leq 0}, \mathcal{T}_{\mathcal{U}}^{\geq 0})$ on $D_{\mathrm{Qcoh},Z}(X)$ is equivalent to the standard t-structure.

Preferred equivalence class. On a triangulated category which has a compact generator, we can intrinsically define an equivalence class. Notable examples of triangulated categories with a compact generator are $D_{\mathrm{Qcoh},Z}(X)$ for a Noetherian scheme and $D(R)$ for a ring spectrum. We begin with the definition, and then prove that the two t-structures we are interested in both lie in the preferred equivalence class, hence proving their equivalence.

Definition 1.15 ([Nee24a])**.** Let $\mathcal{T}$ be a triangulated category with a compact generator G. Then, the *preferred equivalence class* of t-structures on $\mathcal{T}$ consists of those t-structures which are equivalent to the t-structure $(\mathcal{T}_G^{\leq 0}, \mathcal{T}_G^{\geq 0})$ compactly generated by G. Note that this is independent of the choice of the compact generator G.

We begin with the following (well-known and easy) lemma, whose proof we just sketch. The interested reader can see [Nee24a, Lemma 3.5] for a complete proof.

Lemma 1.16. *Let X be a Noetherian finite-dimensional*[16] *scheme. Then, for any perfect complex $E \in \mathrm{Perf}(X)^{\geq n}$, the dual $\mathrm{R}\mathcal{H}\mathrm{om}(E, \mathcal{O}_X) \in \mathrm{Perf}(X)^{\leq -n+\dim X}$.*

Proof Sketch. This statement can be checked affine locally, so without loss of generality we can assume $X = \mathrm{Spec}(R)$ for some Noetherian scheme R of finite Krull dimension. Let P be a projective resolution of E by finitely generated projectives. Then, the required result is clear from the fact that $H^n(\sigma^{\leq n}P)$ has a projective resolution of length less than or equal to $\dim(X)$[17]. $\square$

Theorem 1.17. *Let X be a Noetherian finite-dimensional scheme, and Z a closed subset with $(\mathcal{U}, \mathcal{V})$ a bounded t-structure on $\mathrm{Perf}_Z(X)$. Suppose that the standard t-structure on $D_{\mathrm{Qcoh},Z}(X)$ lies in the preferred equivalence class. Then, the induced t-structure $(\mathcal{T}_{\mathcal{U}}^{\leq 0}, \mathcal{T}_{\mathcal{U}}^{\geq 0})$ on $D_{\mathrm{Qcoh},Z}(X)$ lies in the preferred equivalence class.*

Proof. Let G be a compact generator for $D_{\mathrm{Qcoh},Z}(X)$. We need to show that

$$\mathcal{T}_G^{\leq -N} \subseteq \mathcal{T}_{\mathcal{U}}^{\leq 0} \subseteq \mathcal{T}_G^{\leq N}$$

for some $N \geq 0$. One of the containments is easy. As the t-structure $(\mathcal{U}, \mathcal{V})$ is bounded, some positive shift $G[i]$ lies in $\mathcal{U}$, which immediately implies that $\mathcal{T}_G^{\leq -i} \subseteq \mathcal{T}_{\mathcal{U}}^{\leq 0}$.

The other direction is a bit trickier. For any subcategory $\mathcal{A} \subseteq \mathrm{Perf}_Z(X)$, we denote by $\mathcal{A}^\vee$ the dual full subcategory with objects consisting of $\mathrm{R}\mathcal{H}\mathrm{om}(E, \mathcal{O}_X)$ for $E \in \mathcal{A}$. Consider the dual t-structure $(\mathcal{V}^\vee, \mathcal{U}^\vee)$ on $\mathrm{Perf}_Z(X)$[18]. By the previous argument, there exists some positive shift $G[j]$ which lies in $\mathcal{V}^\vee$, and hence $\mathcal{T}_G^{\leq -j} \subseteq \mathcal{T}_{\mathcal{V}^\vee}^{\leq 0}$. Now, by our assumption that the standard t-structure lies in the preferred equivalence class, we get (by increasing j if necessary) that $D_{\mathrm{Qcoh},Z}(X)^{\leq -j} \subseteq \mathcal{T}_{\mathcal{V}^\vee}^{\leq 0}$. By taking perpendiculars and shifting, we get

$$\mathcal{U}^\vee \subseteq (\mathcal{V}^\vee[1])^\perp = \mathcal{T}_{\mathcal{V}^\vee}^{\geq 0} \subseteq D_{\mathrm{Qcoh},Z}(X)^{\geq -j}$$

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[16] Note that this is exactly where finite-dimensionality is required.

[17] This can be shown easily by the Auslander-Buchsbaum formula. More fundamentally, this is a consequence of the finiteness of the *finitistic dimension* of R, an observation that has been vastly generalised [BCM+23].

[18] This t-structure can be seen as a consequence of the self-duality of the definition of a t-structure via the equivalence $\mathcal{R}\mathrm{Hom}(-, \mathcal{O}_X)\colon \mathrm{Perf}_Z(X)^{\mathrm{op}} \to \mathrm{Perf}_Z(X)$.

Taking the dual again, we get that $\mathcal{U} = (\mathcal{U}^\vee)^\vee \subseteq D_{\mathrm{Qcoh},Z}(X)^{\leq j+\dim X}$ by Lemma 1.16. Again, as the standard t-structure lies in the preferred equivalence class by our hypothesis, we are done. □

Only the last piece of the puzzle remains, which is to show that the standard t-structure lies in the preferred equivalence class of t-structures on $D_{\mathrm{Qcoh},Z}(X)$. For that, we must discuss approximable triangulated categories. We will give some (analytic) motivation for approximability once we give the definition. But first, we begin with some notation.

Notation 1.18. Let $\mathcal{T}$ be a triangulated category with an object G, and let $p \leq q$ and r be integers. Then, we define

- $\langle G\rangle^{[p,q]}$ to be the full subcategory generated by $\{G[-i]\}_{i=p}^{q}$ using all summands, finite coproducts, and extensions,
- $\langle G\rangle^{[p,q]}_r$ to be the full subcategory generated by $\{G[-i]\}_{i=p}^{q}$ using all summands, finite coproducts, and at most $r-1$ extensions.

If $\mathcal{T}$ has all (small) coproducts, then we also define

- $\overline{\langle G\rangle}^{[p,q]}$ to be the full subcategory generated by $\{G[-i]\}_{i=p}^{q}$ using all summands, coproducts, and extensions,
- $\overline{\langle G\rangle}^{[p,q]}_r$ to be the full subcategory generated by $\{G[-i]\}_{i=p}^{q}$ using all summands, coproducts, and at most $r-1$ extensions.

Definition 1.19. A triangulated category $\mathcal{T}$ is *approximable* (resp. *weakly approximable*) if it has a single compact generator G, and a t-structure $(\mathcal{T}^{\leq 0}, \mathcal{T}^{\geq 0})$ such that for some positive integer N

(1) $G[N] \in \mathcal{T}^{\leq 0}$,
(2) $\mathrm{Hom}(G, \mathcal{T}^{\leq -N}) = 0$,
(3) for all $F \in \mathcal{T}^{\leq 0}$, there exists a triangle $E \to F \to D \to E[1]$ with $E \in \overline{\langle G\rangle}^{[-N,N]}_N$ (resp. $\overline{\langle G\rangle}^{[-N,N]}$) and $D \in \mathcal{T}^{\leq -1}$.

For the purpose of these notes, we will call a t-structure satisfying the above definition for some (equivalently all) compact generator, and some positive integer N *approximating* (resp. *weakly approximating*).

The relevance of this definition to our problem is the following lemma.

Lemma 1.20. *Any (weakly) approximating t-structure lies in the preferred equivalence class (Definition 1.15). If $\mathcal{T}$ is (weakly) approximable then any t-structure in the preferred equivalence class is (weakly) approximating.*

Proof sketch. Let $(\mathcal{T}^{\leq 0}, \mathcal{T}^{\geq 0})$ be weakly approximating, and let $F \in \mathcal{T}^{\leq 0}$. So, $(\mathcal{T}^{\leq 0}, \mathcal{T}^{\geq 0})$ satisfies the conditions of Definition 1.19 for some compact generator G and some positive integer N. We need to show that the t-structure $(\mathcal{T}^{\leq 0}, \mathcal{T}^{\geq 0})$ is equivalent to the t-structure $(\mathcal{T}^{\leq 0}_G, \mathcal{T}^{\geq 0}_G)$ generated by G. As $G[N] \in \mathcal{T}^{\leq 0}$, we immediately get[19] that $\mathcal{T}^{\leq -N}_G = \mathrm{Coprod}(\{G[i]\}_{i\geq N}) \subseteq \mathcal{T}^{\leq 0}$. What remains to show is that $\mathcal{T}^{\leq 0} \subseteq \mathcal{T}^{\leq i}_G$ for some integer i.

Let $F \in \mathcal{T}^{\leq 0}$. Inductively we can show the existence of a sequence $E_1 \to E_2 \to \cdots$ with compatible maps $f_i\colon E_i \to F$ such that in the triangles $E_n \xrightarrow{f_i} F \to D_n \to E_n[1]$ we have

[19] This uses the fact that $\mathcal{T}^{\leq 0}$ is closed under coproducts, as can be seen from the equality $\mathcal{T}^{\leq 0} = {}^{\perp}\mathcal{T}^{\geq 1}$.

that $E_n \in \overline{\langle G \rangle}^{[-N-n,N]}$ and $D_n \in \mathcal{T}^{\leq -n-1}$. From this, as shown in Idea 1.22, we get the isomorphism $\underrightarrow{\mathrm{hocolim}}\, E_i \to F$. As

$$\overline{\langle G \rangle}^{[-N-n,N]} \subseteq \mathrm{Coprod}(\{G[i]\}_{i \geq -N}) = \mathcal{T}_G^{\leq N},$$

for all $n \geq 0$, we have that $F \cong \underrightarrow{\mathrm{hocolim}}\, E_i \in \mathcal{T}_G^{\leq N}$ which is what we needed to show. □

Recall that what remained to show is that the standard t-structure on $D_{\mathrm{Qcoh},Z}(X)$ lies in the preferred equivalence class. In light of the above lemma, this is what the following result shows.

Theorem 1.21 ([Nee24a]). *Let X be a Noetherian scheme and Z a closed subset. Then*

(1) the standard t-structure on $D_{\mathrm{Qcoh},Z}(X)$ is weakly approximating,

(2) if X is separated then the standard t-structure on $D_{\mathrm{Qcoh}}(X)$ is approximating.

This result is of central importance, as it shows that abstract results proven using approximability are guaranteed to have applications to algebraic geometry. We will not have the time to delve into the proof of this result in this generality, but we will see proofs of this result in some special situations at different points in the course. We should mention that there are triangulated categories arising in homotopy theory and representation theory which are also approximable. We will see a large class of examples in Proposition 1.24. Before we get to that, let us take a closer look at the notion of (weak) approximability.

What is approximability?

Idea 1.22. There are many results in analysis on approximating complicated objects (often these objects are functions) by simpler ones. Important examples of these are the approximation of continuous function on closed intervals of $\mathbb{R}$ (Stone-Weierstrass theorem) and the approximation of square integrable functions on the circle by the partial sums of its Fourier series. We draw an analogy of approximability with the latter.

Let $\mathcal{T}$ be an approximable triangulated category with $(\mathcal{T}^{\leq 0}, \mathcal{T}^{\geq 0})$ an approximating t-structure. For any object $F \in \mathcal{T}^{\leq i}$, one can inductively find a sequence $F_1 \to F_2 \to \cdots$ mapping to F such that in the triangle $F_n \to F \to D_n \to F_n[1]$, we have $F_n \in \overline{\langle G \rangle}_{nN}^{[-n-N,N+i]}$ and $D_n \in \mathcal{T}^{\leq -n-1}$. This gives us a map $\underrightarrow{\mathrm{hocolim}}\, F_n \to F$. By Definition 1.19(2), for every integer i, we have that $\mathrm{Hom}(G[i], D_n) = 0$ for $n \gg 0$. And so for every integer i, we get an isomorphism $\mathrm{Hom}(G[i], F_n) \to \mathrm{Hom}(G[i], F)$ for $n \gg 0$. Finally, this gives us that the induced map

$$\mathrm{Hom}(G[i], \underrightarrow{\mathrm{hocolim}}\, F_n) \simeq \underrightarrow{\lim}\, \mathrm{Hom}(G[i], F_n) \to \mathrm{Hom}(G[i], F)$$

is an isomorphism, and as G is a compact generator, we get that $\underrightarrow{\mathrm{hocolim}}\, F_n \to F$ is an isomorphism.

On the other hand, let $\mathbb{T}$ denote the circle, and $\mathcal{M}(\mathbb{T})$ and $L^2(\mathbb{T})$ the measurable and the square integrable functions on the circle respectively. Finally, let $\| - \|_2$ denote the L^2 norm and $S_n(f)$ the partial sum $\Sigma_{-n}^{n} \hat{f}(n) e^{nx}$.

Analysis	Approximability
$\mathcal{M}(\mathbb{T})$	$\mathcal{T}$
$\{e^{nx}\}$	$\{G[n]\}$
If $\langle e^{nx}, f\rangle = 0$ for all $n \in \mathbb{Z}$ then $f = 0$.	If $\mathrm{Hom}(G[n], F) = 0$ for all $n \in \mathbb{Z}$ then $F = 0$.
$L^2(\mathbb{T})$	$\mathcal{T}^- := \bigcup_{i\geq 0} \mathcal{T}^{\leq i}$
$\{f : \\|f\\|_2 < 1/2^n\}$	$\mathcal{T}^{\leq -n-1}$
For $f \in L^2(\mathbb{T})$ there exists $\{S_n(f)\}$, built from $\{e^{ix}\}$ "in n-steps".	For $F \in \mathcal{T}^-$ there exists $\{F_n\}$, built from $\{G[i]\}$ in "in n-steps".
$\\|S_n(f) - f\\|_2$ goes to zero.	$\mathrm{Cone}(F_n \to F) \in \mathcal{T}^{\leq -n-1}$.
$S_N(f)$ converges to f.	$\underrightarrow{\mathrm{hocolim}}\, F_n \xrightarrow{\sim} F$.

We now look at the instructive (and simple) case of affine schemes where we can see the analogy of Idea 1.22 in action.

Example 1.23. Let R be a (not necessarily commutative) ring. The object R is a compact generator of the derived category of (right) R-modules. We now show that the standard t-structure $(\mathbf{D}(R)^{\leq 0}, \mathbf{D}(R)^{\geq 0})$ is approximating. Consider a complex F in $\mathbf{D}(R)^{\leq 0}$. Then, up to quasi-isomorphism, we can write the complex as,

$$\cdots \to P^{-n-2} \to P^{-n-1} \to P^{-n} \to \cdots \to P^{-1} \to P^0 \to 0 \to \cdots$$

with P^{-i} projective modules for all $i \geq 0$. We get the following commutative diagram for any $n \geq 0$,

$$\begin{tikzcd}
\cdots \arrow[r] & 0 \arrow[r] \arrow[d] & 0 \arrow[r] \arrow[d] & P^{-n} \arrow[r] \arrow[d] & \cdots \arrow[r] & P^{-1} \arrow[r] \arrow[d] & P^0 \arrow[r] \arrow[d] & 0 \arrow[r] \arrow[d] & \cdots \\
\cdots \arrow[r] & P^{-n-2} \arrow[r] \arrow[d] & P^{-n-1} \arrow[r] \arrow[d] & P^{-n} \arrow[r] \arrow[d] & \cdots \arrow[r] & P^{-1} \arrow[r] \arrow[d] & P^0 \arrow[r] \arrow[d] & 0 \arrow[r] \arrow[d] & \cdots \\
\cdots \arrow[r] & P^{-n-2} \arrow[r] & P^{-n-1} \arrow[r] & 0 \arrow[r] & \cdots \arrow[r] & 0 \arrow[r] & 0 \arrow[r] & 0 \arrow[r] & \cdots
\end{tikzcd}$$

This gives us maps of complexes, $E_n \to F \to D_n$, with E_n a bounded complex of projectives, and D_n with vanishing cohomology for all degrees greater than $-n$. As the maps form a degree-wise split exact sequence, $E_n \to F \to D_n$ extends to a distinguished triangle, $E_n \to F \to D_n \to E_n[1]$. The complexes D_n represent "error terms" of sorts. They get smaller and smaller, in the sense that $D_{n+1} \in \mathbf{D}(R)^{\leq -n-2} \subseteq \mathbf{D}(R)^{\leq -n-1}$.

What remains to be shown is that the complexes E_n are built from the compact generator R in finitely many steps, which we will do by an induction on n. Note that the object E_n is a complex of projective modules in degrees $-n$ to 0. For the base case, let P be any projective module. Then, it is a summand of a free module, that is, it is built from R by direct sums and summands. Now, for any complex E_{n+1} of projective modules in degrees $-n-1$ to 0, we have the following commutative diagram,

$$\begin{tikzcd}
\cdots \arrow[r] & 0 \arrow[r] \arrow[d] & 0 \arrow[r] \arrow[d] & P^n \arrow[r] \arrow[d] & \cdots \arrow[r] & P^0 \arrow[r] \arrow[d] & 0 \arrow[r] \arrow[d] & \cdots \\
\cdots \arrow[r] & 0 \arrow[r] \arrow[d] & P^{-n-1} \arrow[r] \arrow[d] & P^n \arrow[r] \arrow[d] & \cdots \arrow[r] & P^0 \arrow[r] \arrow[d] & 0 \arrow[r] \arrow[d] & \cdots \\
\cdots \arrow[r] & 0 \arrow[r] & P^{-n-1} \arrow[r] & 0 \arrow[r] & \cdots \arrow[r] & 0 \arrow[r] & 0 \arrow[r] & \cdots
\end{tikzcd}$$

Again, this is degree-wise split, so it extends to a triangle $E_n \to E_{n+1} \to P^{-n-1}[n+1] \to E_n[1]$, where E_n is a complex of projective modules in degrees $-n$ to 0. So, inductively we get that E_n can be constructed from R using direct sums, summands and at most n shifts and n extensions. This is what we mean by saying that an object is "built in finitely many steps from the compact generator R". Note that the argument above for $n = 1$ proves that the standard t-structure is approximating.

Proposition 1.24. *Let $\mathcal{T}$ be a triangulated category. If G is a compact generator such that*

(1) $\operatorname{Hom}(G, G[n]) = 0$ *for all $n \geq 1$ then $\mathcal{T}$ is approximable,*

(2) $\operatorname{Hom}(G, G[n]) = 0$ *for all $n \geq 2$ then $\mathcal{T}$ is weakly approximable.*

Proof. We will prove that the t-structure $(\mathcal{T}_G^{\leq 0}, \mathcal{T}_G^{\geq 0})$ generated by G (Notation 1.10) is approximating (resp. weakly approximating). The first two conditions of Definition 1.19 are easily seen to hold in both the cases. Recall that $\mathcal{T}_G^{\leq 0} = \operatorname{Coprod}(\{G[i]\}_{i\geq 0})$. So the first (resp. second) hypothesis along with the fact that G is compact implies that $\operatorname{Hom}(G, \mathcal{T}_G^{\leq -i}) = 0$ for all $i \geq 1$ (resp. $i \geq 2$).

What remains to be shown is the third condition of Definition 1.19. Let us begin with the proof when the first hypothesis holds. Let $F \in \mathcal{T}_G^{\leq 0}$. We define $E = \bigoplus_{\operatorname{Hom}(G,F)} G$, and let $E \to F \to D \to E[1]$ be a triangle defined by the universal property of coproducts. As $E \in \overline{\langle G \rangle}_1^{[0,0]} \subseteq \overline{\langle G \rangle}_1^{[-1,1]}$, it is enough to show that $D \in \mathcal{T}_G^{\leq -1}$. This is equivalent to showing that in the truncation triangle $D^{\leq -1} \to D \to D^{\geq 0} \to D^{\leq -1}[1]$, we have that $D^{\geq 0} = 0$. Further, as G is a compact generator, we just have to prove the vanishing of $\operatorname{Hom}(G[n], D^{\geq 0})$ for all $n \in \mathbb{Z}$. Note that $\operatorname{Hom}(G[n], D^{\geq 0}) = 0$ for all $n \geq 1$ as $D^{\geq 0} \in \mathcal{T}_G^{\geq 0}$, so we just need to show the vanishing for non-positive integers n.

Towards that end, let $n \leq 0$, and $f \colon G[n] \to D^{\geq 0}$ be any morphism. As the composite $G[n] \xrightarrow{f} D^{\geq 0} \to D^{\leq -1}[1]$ vanishes[20], f factors as $G[n] \xrightarrow{g} D \to D^{\geq 0}$. Note that the composite $G[n] \xrightarrow{g} D \to E[1] = \bigoplus_{\operatorname{Hom}(G,F)} G[1]$ also vanishes by our hypothesis. So, g factors as $G[n] \xrightarrow{h} F \to D$. But by definition, it is clear that h factors via $E \to F$, and hence g and f factor via the composite $E \to F \to D$. As consecutive arrows in a triangle compose to zero, we are done.

Now assume that the second hypothesis holds. Define the full subcategory

$$\mathcal{S} := \{F : \exists \text{ triangle } E \to F \to D \to E[1] \text{ with } E \in \overline{\langle G \rangle}^{[-0,0]}, D \in \mathcal{T}_G^{\leq -1}\}.$$

It is clear that $\mathcal{S}$ is closed under coproducts, and that $\{G[i]\}_{i\geq 0} \subseteq \mathcal{S}$. Therefore, if $\mathcal{S}$ is also closed under taking extensions, then $\mathcal{T}_G^{\leq 0} = \operatorname{Coprod}(\{G[i]\}_{i\geq 1}) \subseteq \mathcal{S}$ which is what we needed to show.

[20]Again, by the argument from the first paragraph of this proof.

So, we need to show that for any triangle $F_1 \to F \to F_2 \to F_1[1]$ with $F_1, F_2 \in \mathcal{S}$ we have that F also lies in $\mathcal{S}$. As $F_1, F_2 \in \mathcal{S}$, we have triangles $E_i \to F_i \to D_i \to E_i[1]$ with $E_i \in \overline{\langle G \rangle}^{[-0,0]}$ and $D_i \in \mathcal{T}_G^{\leq -1}$ for $i \in \{1, 2\}$, giving us the diagram

$$\begin{tikzcd}
E_1 \arrow[d] & & E_2 \arrow[d] & E_1[1] \arrow[d] \\
F_1 \arrow[r] \arrow[d] & F \arrow[r] & F_2 \arrow[r] \arrow[d] & F_1[1] \arrow[d] \\
D_1 & & D_2 & D_1[1]
\end{tikzcd}$$

Note that $\mathrm{Hom}(\overline{\langle G \rangle}^{[-0,0]}, \mathcal{T}_G^{\leq -2}) = 0$ by our hypothesis and the compactness of G. So, the composite $E_2 \to F_2 \to F_1[1] \to D_2[1]$ vanishes as $E_2 \in \overline{\langle G \rangle}^{[-0,0]}$ and $D_2[1] \in \mathcal{T}_G^{\leq -2}$. Therefore, we can complete the above diagram to a morphism of triangles

$$\begin{tikzcd}
E_1 \arrow[r] \arrow[d] & E \arrow[r] \arrow[d] & E_2 \arrow[r] \arrow[d] & E_1[1] \arrow[d] \\
F_1 \arrow[r] \arrow[d] & F \arrow[r] \arrow[d] & F_2 \arrow[r] \arrow[d] & F_1[1] \arrow[d] \\
D_1 \arrow[r] & D \arrow[r] & D_2 \arrow[r] & D_1[1]
\end{tikzcd}$$

which gives us the required triangle $E \to F \to D \to E[1]$. $\square$

Remark 1.25. Proposition 1.24 implies that $D(R)$ is approximable for any connective DG-algebra/ring spectrum R.

2. Strong generators and regularity

Let us begin with some notation and the definition of strong generation.

Notation 2.1. Let $\mathcal{T}$ be a triangulate category and $G \in \mathcal{T}$. Then

- for any $n \geq 1$, $\langle G \rangle_n$ is the full subcategory generated by G using all shifts, summands, finite coproducts, and at most $n-1$ extensions. Note that $\langle G \rangle_{n+1} = \langle G \rangle_n * \langle G \rangle_1 = \langle G \rangle_1 * \langle G \rangle_n$.
- $\langle G \rangle$ is the thick subcategory generated by G, that is, the triangulated subcategory closed under summands generated by G. Note that $\langle G \rangle = \bigcup_{n \geq 1} \langle G \rangle_n = \bigcup_{n \geq 1} \langle G \rangle_n^{[-n,n]}$.

If $\mathcal{T}$ has coproducts, then we can also define,

- for any $n \geq 1$, $\langle G \rangle_n$ is the full subcategory generated by G using all shifts, summands, coproducts, and at most $n-1$ extensions.
- $\overline{\langle G \rangle}_n$ is the localising subcategory generated by G, that is, the triangulated subcategory closed under coproducts (and hence also summands) generated by G.

Definition 2.2. Let $\mathcal{T}$ an essentially small triangulated category. An object $G \in \mathcal{T}$ is a

(1) *classical generator* if every object in $\mathcal{T}$ can be built from G using summands, finite coproducts, shifts, and extensions. That is, $\mathcal{T} = \langle G \rangle$.

(2) *strong generator* if there exists a positive integer n such that every object in $\mathcal{T}$ can be built from G using summands, finite coproducts, shifts, and at most $n-1$ extensions. That is, $\mathcal{T} = \langle G \rangle_n$.

It is clear than any strong generator is a classical generator. Further, if $\mathcal{T}$ is strongly generated then every classical generator is a strong generator.

We will see that there is the existence of a strong generator is closely related to regularity for Noetherian schemes. This is made precise in the following result by Neeman, which settles a conjecture from [BVdB03] where a similar result was proven for smooth schemes over a field.

Theorem 2.3 ([Nee21c])**.** *Let X be a Noetherian, separated scheme. Then,* $\operatorname{Perf}(X)$ *has a strong generator if and only if X is regular and finite-dimensional.*

The existence of a strong generator implies many pleasant consequences, perhaps the most important of which is the following representability theorem.

Theorem 2.4 ([BVdB03]&[Rou08])**.** *Let $\mathcal{T}$ be R-linear triangulated category for a Noetherian commutative ring such that $\bigoplus_{i\in\mathbb{Z}} \operatorname{Hom}(X, Y[i])$ is a finitely generated R module for all $X, Y \in \mathcal{T}$. If $\mathcal{T}$ has a strong generator then the following are equivalent for a (co)homological functor*[21] *$H : \mathcal{T} \to \operatorname{Mod}(R)$*

(1) H is representable,
(2) H is a finite (co)homological functor*, that is $\bigoplus_{i\in\mathbb{Z}} H(E[i])$ is a finitely generated R-module for all $E \in \mathcal{T}$.*

This result has many applications in algebraic geometry. We mention a few of them in the following corollary.

Corollary 2.5. *Let X, Y be regular schemes which are proper over a Noetherian affine scheme. Then*

(1) Any left (or right) admissible subcategory of $D^b_{\mathrm{coh}}(X)$ is admissible.
(2) Let $\Phi : D_{\mathrm{Qcoh}}(X) \leftrightarrows D_{\mathrm{Qcoh}}(Y) : \Psi$ be adjoint functors. Then, Φ restricts to $D^b_{\mathrm{coh}}(-)$ if and only if Ψ does.

Let us now get back to the connection between regularity and strong generation and prove Theorem 2.3 for the affine situation. But first, we recall (a version of) the Ghost lemma.

Lemma 2.6. *Let $\mathcal{T}$ be a triangulated category and $G \in \mathcal{T}$. Let $f : X \to Y$ be a map such that*

(1) $f = f_1 \cdots f_n$,
(2) $\operatorname{Hom}(G[j], f_i) = 0$ for all $j \in \mathbb{Z}$[22]*and $1 \le i \le n$,*
(3) $X \in \langle G \rangle_n$.

Then, $f = 0$.

Proof. We prove this inductively on $n \ge 1$. The base case is clear. So now we assume that conditions (1)-(3) hold for some $n \ge 2$. As $X \in \langle G \rangle_n = \langle G \rangle_{n-1} * \langle G \rangle_1$, we have a triangle $X' \xrightarrow{g} X \xrightarrow{h} X'' \to X'[1]$ with $X' \in \langle G \rangle_{n-1}$ and $X'' \in \langle G \rangle_1$. Then, the map $f_{n-1} \circ f_{n-2} \cdots (f_1 \circ g) = 0$ by the induction hypothesis. Therefore $f_{n-1} \cdots f_1 = p \circ h$ for some map p. Finally, by the base case $f = f_n \cdots f_1 = f_n \circ p \circ h = 0$ by the case $n = 1$. The argument

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22 Such maps are called *G-ghosts*.

is perhaps clearer by looking at the diagram below, in which the arrows coloured red vanish by the induction hypothesis.

$$\begin{tikzcd}
X' \arrow[r, "g"] \arrow[ddd, bend right, red, "0=f_{n-1}\cdots f_1\circ g"'] & X \arrow[r, "h"] \arrow[d, "f_1"'] & X'' \arrow[ddl, dashed, "p"] \arrow[dddd, bend left, red, "0=f_n\circ p"] \\
& Y_1 \arrow[d] & \\
& \vdots \arrow[d, "f_{n-1}"'] & \\
& Y_{n-1} \arrow[d, "f_n"] & \\
& Y_n &
\end{tikzcd}$$

□

Theorem 2.7. *Let R be a Noetherian ring. Then the following are equivalent*

(1) R is regular and finite-dimensional.
(2) R has finite global dimension.
(3) $\mathrm{Perf}(R)$ has a strong generator.

Proof sketch. (1) $\iff$ (2) is well known commutative algebra which has already been used above. For (2) $\implies$ (3) note that if global dimension of R is n then $\mathrm{Perf}(R) \xrightarrow{\sim} D^b(\mathrm{mod}(R))$ and that every module has a projective resolution of length less than or equal to n. The latter immediately implies that any finitely generated module lies in $\langle R\rangle_{n+1}$. As any complex can be written as an extensions of the kernels and the images of the involved differentials we get that $D^b(\mathrm{mod}(R)) = \langle R\rangle_{2(n+1)}$[23].

Finally, we show (3) $\implies$ (2). Note that for any $F \in D^b(\mathrm{mod}(R))$, the functor $\mathrm{Hom}(-, F)|_{\mathrm{Perf}(R)}$ is a finite cohomological functor. So, by Theorem 2.4 and the Yoneda lemma, there is a perfect complex E along with a map $E \to F$ inducing an isomorphism $\mathrm{Hom}(-, E)|_{\mathrm{Perf}(R)} \xrightarrow{\sim} \mathrm{Hom}(-, F)|_{\mathrm{Perf}(R)}$. This in turn implies that the map $E \to F$ is an isomorphism, which shows us that R is regular.

As $\mathrm{Perf}(R)$ is strongly generated, and R is a classical generator, there exists $n \geq 1$ such that $\mathrm{Perf}(R) = \langle R\rangle_n$. Let M be any finitely generated R-module. By the regularity of R, there exists a projective resolution P by finitely generated projectives of finite length. Note that the map $P = \sigma^{\leq 0}P \xrightarrow{f} \sigma^{\leq -n}P$ to the brutal truncation at the n^{th} stage factors as $\sigma^{\leq 0}P \xrightarrow{f_1} \sigma^{\leq -1}P \xrightarrow{f_2} \cdots \xrightarrow{f_n} \sigma^{\leq -n-1}P$. It is easy to see that

$$\mathrm{Hom}(R[-i], f_j) = H^i(\sigma^{\leq -j+1}) \xrightarrow{H^i(f_j)} H^i(\sigma^{\leq -j}) = 0$$

for all $1 \leq j \leq n$ as $H^i(\sigma^{\leq -j}) = 0$ for all $i \neq j$. So, $f = 0$ by Lemma 2.6, and hence M is a summand of $\sigma^{\geq -n}P$ implying that its projective dimension is less than or equal to $n-1$. As M was arbitrary, the global dimension must also be bounded above by $n-1$, and we are done. □

[23] Using a slightly more sophisticated argument, we can bring the number of extensions down to n, see [Nee21b, Theorem 7.5].

Remark 2.8. Suppose $\mathrm{Perf}(X)$ has a strong generator for a Noetherian, separated scheme X. Let U be any affine open subscheme. As $\mathrm{Perf}(U)$ is a Verdier quotient of $\mathrm{Perf}(X)$, we get that X is regular and finite-dimensional by an immediate application of Theorem 2.7.

Conversely, suppose X is regular and finite-dimensional. By Theorem 2.7, it has an open affine cover where $\mathrm{Perf}(U)$ has a strong generator. So, what remains to show is that the existence of strong generators for the perfect complexes can be checked (Zariski) locally.

To prove this, we need to introduce the following notion[24].

Definition 2.9. A map $f\colon X \to Y$ of Noetherian schemes is said to be compactly (resp. coherently) bounded if for all F in $\mathrm{Perf}(X)$ (resp. $D^b_{\mathrm{coh}}(X)$), there exists an object G in $\mathrm{Perf}(Y)$ (resp. $D^b_{\mathrm{coh}}(Y)$) and an positive integer n such that $Rf_*F \in \overline{\langle G \rangle}_n$.

The following lemma shows that open immersions are compactly bounded.

Lemma 2.10. *Any open immersion $f\colon V \to X$ of separated Noetherian schemes is compactly bounded. In fact*

(1) $\mathrm{Hom}(Rf_*\mathcal{O}_V, D_{\mathrm{Qcoh}}(X)^{\leq -n}) = 0$ *for* $n \gg 0$,
(2) for any compact object $H \in \mathrm{Perf}(V)$ and any compact generator G of $D_{\mathrm{Qcoh}}(X)$, there exists $L > 0$ such that $Rf_*H \in \overline{\langle G \rangle}_L$.

Proof. We first prove that (1) $\implies$ (2). So, assume that

$$\mathrm{Hom}(Rf_*\mathcal{O}_V, D_{\mathrm{Qcoh}}(X)^{\leq -n}) = 0 \text{ for } n \gg 0$$

Let $H \in \mathrm{Perf}(U)$ be any object. By Thomason's localisation theorem[25], there is a perfect complex $\widetilde{H} \in \mathrm{Perf}(X)$ such that H is a summand of $Lf^*\widetilde{H}$. So, Rf_*H is a summand of $Rf_*Lf^*\widetilde{H} = \widetilde{H} \otimes Rf_*\mathcal{O}_V$ by projection formula. But,

$$\mathrm{Hom}(\widetilde{H} \otimes Rf_*\mathcal{O}_V, D_{\mathrm{Qcoh}}(X)^{\leq -n}) = \mathrm{Hom}(Rf_*\mathcal{O}_V, \widetilde{H}^\vee \otimes D_{\mathrm{Qcoh}}(X)^{\leq -n}) = 0 \text{ for } n \gg 0$$

by our assumption. And so,

$$\mathrm{Hom}(Rf_*H, D_{\mathrm{Qcoh}}(X)^{\leq -n}) = 0 \text{ for } n \gg 0.$$

By approximability we have triangles $E_n \to Rf_*H \to D_n \to E_n[1]$ with $E \in \overline{\langle G \rangle}_{Nn}$ and $D_n \in D_{\mathrm{Qcoh}}(X)^{\leq -n-1}$ for some fixed compact generator G and positive integer N. So, by the vanishing above, we get that Rf_*H is a summand of E_n for some $n \gg 0$, and hence lies in $\overline{\langle G \rangle}_{Nn}$, proving (2).

What remains to prove is (1), which can be done by an affine local computation of the projective dimension of pushforward of the structure sheaf. Let $X = \bigcup_{i=1}^n U_i$ be a cover by affine opens. Then the map f factors as

$$V \to V \cup U_1 \to \cdots \to V \cup \bigcup_{i=1}^n U_i = X.$$

[24] Although we state it for schemes only, see [DLM25] for a more general definition.
[25] See [?]. The result

So, by an inductive argument, we may as well assume that $X = V \cup U$ for an affine scheme U. We get a diagram,

$$\begin{tikzcd} & U = \mathrm{Spec}(R) \arrow[dr, "h"] & \\ W = U \cap V \arrow[ur, "g"] \arrow[dr, "e"'] & & X \\ & V \arrow[ur, "f"'] & \end{tikzcd}$$

Considering the triangle

$$Q \to \mathcal{O}_X \to Rf_*\mathcal{O}_V \to Q[1]$$

we get from the unit of adjunction. By the corresponding long exact sequence, it is enough to prove that

$$\mathrm{Hom}(Q, D_{\mathrm{Qcoh}}(X)^{\leq -n}) = 0 \text{ for } n \gg 0.$$

As $\mathcal{O}_X \to Rf_*\mathcal{O}_V$ is an isomorphism on the open subscheme $V \subseteq X$, $Q \in D_{\mathrm{Qcoh}, X\setminus V}(X)$. But $X \setminus V \subseteq U$, and hence $Q = Rh_*Lh^*Q$ via the equivalence

$$Lh^* \colon D_{\mathrm{Qcoh}, X\setminus V}(X) \leftrightarrows D_{\mathrm{Qcoh}, U\setminus V}(X) : Rh_*.$$

Applying Lh^* to the triangle above and using flat base change, we get

$$\widetilde{Q} \to \mathcal{O}_U \to Rg_*\mathcal{O}_W \to \widetilde{Q}[1]$$

where $\widetilde{Q} := Lh^*Q$ and the map $\mathcal{O}_U \to Rg_*\mathcal{O}_W$ is the unit of adjunction. Note that as $U = \mathrm{Spec}(R)$, there is a very concrete description of $Rg_*\mathcal{O}_W$ via the Čech complex and that of $\widetilde{Q}$ via local cohomology. In particular, if the ideal defined by $U \setminus W$ is generated by $r_1, \cdots r_s \in R$, then $\widetilde{Q} = \bigotimes_{i=1}^{s}(\mathrm{Cone}(R \to R_{r_i})[-1])$. But, $\mathrm{Cone}(R \to R_r) = \underrightarrow{\mathrm{hocolim}}(\mathrm{Cone}(R \xrightarrow{r^n} R))$. So, by the triangle defining homotopy colimit, we get that $\widetilde{Q}$ has a projective resolution which is zero outside degrees $-1 \leq i \leq s$.

Now, let $F \in D_{\mathrm{Qcoh}}(X)^{\leq -2}$. We have the Mayer-Vietoris triangle (see for example [Sta24, Tag 08BV])

$$F \to Rh_*F|_U \oplus Rg_*F|_V \to R(fe)_*F|_W \to F[1].$$

Note that $\mathrm{Hom}(Q, Rf_*D_{\mathrm{Qcoh}}(X)) = 0$, and hence by the long exact sequence we get by the above triangle we get that

$$\mathrm{Hom}(Q, F) = \mathrm{Hom}(Q, Rg_*Lg^*F) = \mathrm{Hom}(Lg^*Q, Lq^*F) = 0$$

where the final equality holds from the bound on the projective resolution of $Lg^* = \widetilde{Q}$ calculated above.

□

Finally, we prove our main theorem using the compact boundedness of open immersions.

Theorem 2.11. *Let X be a Noetherian, separated scheme. Then, the following are equivalent:*

(1) X is regular and finite-dimensional.
(2) X has an open cover $X = \bigcup_1^n U_i$ such that $\mathrm{Perf}(U_i)$ is strongly generated,
(3) $\mathrm{Perf}(X)$ is strongly generated.

Proof. (1) $\iff$ (2) and (3) $\implies$ (2) is shown in Remark 2.8. What remains to show is that (2) $\implies$ (3). We prove this result by an induction on n. If $n = 1$, then the result trivially holds.

So, assume that X has an open cover $X = \bigcup_1^n U_i$ such that $\mathrm{Perf}(U_i)$ is strongly generated for some $n \geq 2$. We can write $X = U \cup V$ where $U = U_1$ and $V = \bigcup_2^n U_i$. By induction hypothesis, both $\mathrm{Perf}(V)$ and $\mathrm{Perf}(U \cap V)$ are strongly generated, the latter as $U \cap V = \bigcup_2^n (U_1 \cap U_i)$. By the Mayer-Vietoris triangle ([Sta24, Tag 08BV]), we have that

$$\mathrm{Perf}(X) \subseteq (Rj_{U,*}\,\mathrm{Perf}(U) \oplus Rj_{V,*}\,\mathrm{Perf}(V)) * Rj_{U\cap V,*}\,\mathrm{Perf}(U \cap V))$$

where j_U, j_V, and $j_{U\cap V}$ are the open immersions corresponding to U, V, and $U \cap V$ respectively.

From this, using the compact boundedness of open immersions, we get that $\mathrm{Perf}(X) \subseteq \overline{\langle G\rangle}_N$ for some positive integer N and compact generator G. Finally, this implies that $\mathrm{Perf}(X) = \langle G\rangle_N$ as G is a compact object, see [Nee21c, Proposition 1.9]. □

References

[AGH19] Benjamin Antieau, David Gepner, and Jeremiah Heller. K-theoretic obstructions to bounded t-structures. *Invent. Math.*, 216(1):241–300, 2019.

[ATJLS03] Leovigildo Alonso Tarrío, Ana Jeremías López, and María José Souto Salorio. Construction of t-structures and equivalences of derived categories. *Trans. Am. Math. Soc.*, 355(6):2523–2543, 2003.

[BCM+23] Rudradip Biswas, Hongxing Chen, Kabeer Manali Rahul, Chris J. Parker, and Junhua Zheng. Bounded t-structures, finitistic dimensions, and singularity categories of triangulated categories. arXiv:2401.00130, 2023.

[BL26] Burklund and Levy. Some aspects of noncommutative geometry. Preprint available here, cited version accessed in April 2026.

[BNP23] Jesse Burke, Amnon Neeman, and Bregje Pauwels. Gluing approximable triangulated categories. *Forum Math. Sigma*, 11:18, 2023. Id/No e110.

[BVdB03] A. Bondal and M. Van den Bergh. Generators and representability of functors in commutative and noncommutative geometry. *Mosc. Math. J.*, 3(1):1–36, 2003.

[CG24] Charley Cummings and Sira Gratz. Metric completions of discrete cluster categories. arXiv:2407.17369, 2024.

[CHNS24] Alberto Canonaco, Christian Haesemeyer, Amnon Neeman, and Paolo Stellari. The passage among the subcategories of weakly approximable triangulated categories. arXiv:2402.04605, 2024.

[CNS25] Alberto Canonaco, Amnon Neeman, and Paolo Stellari. Weakly approximable triangulated categories and enhancements: a survey. *Boll. Unione Mat. Ital.*, 18(1):109–134, 2025.

[DLM25] Timothy De Deyn, Pat Lank, and Kabeer Manali Rahul. Descending strong generation in algebraic geometry. Preprint, arXiv:2502.08629 [math.AG] (2025), 2025.

[GM03] S. I. Gelfand and Yu. I. Manin. *Methods of homological algebra. Transl. from the Russian.* Springer Monogr. Math. Berlin: Springer, 2nd ed. edition, 2003.

[IT19] Srikanth B. Iyengar and Ryo Takahashi. Openness of the regular locus and generators for module categories. *Acta Math. Vietnam.*, 44(1):207–212, 2019.

[LN07] Joseph Lipman and Amnon Neeman. Quasi-perfect scheme-maps and boundedness of the twisted inverse image functor. *Ill. J. Math.*, 51(1):209–236, 2007.

[LS25] Ishan Levy and Vladimir Sosnilo. c-structures and trace methods beyond connective rings. arXiv:2509.14774, 2025.

[Mat26a] Cyril Matoušek. Metric completions of triangulated categories from finite dimensional algebras. arXiv:2409.01828, 2026.

[Mat26b] Cyril Matoušek. Metric completions of triangulated categories from hereditary rings. arXiv:2508.20283, 2026.

[Nee18] Amnon Neeman. The categories $\mathcal{T}^c$ and $\mathcal{T}^b_c$ determine each other. arXiv:1806.06471, 2018.

[Nee20] Amnon Neeman. Metrics on triangulated categories. *J. Pure Appl. Algebra*, 224(4):13, 2020. Id/No 106206.

[Nee21a] Amnon Neeman. Approximable triangulated categories. In *Representations of algebras, geometry and physics, Maurice Auslander distinguished lectures and international conference, Woods Hole*

Oceanographic Institute, Woods Hole, MA, USA, April 25–30, 2018, pages 111–155. Providence, RI: American Mathematical Society (AMS), 2021.

[Nee21b] Amnon Neeman. Approximable triangulated categories. In *Representations of algebras, geometry and physics, Maurice Auslander distinguished lectures and international conference, Woods Hole Oceanographic Institute, Woods Hole, MA, USA, April 25–30, 2018*, pages 111–155. Providence, RI: American Mathematical Society (AMS), 2021.

[Nee21c] Amnon Neeman. Strong generators in $\mathbf{D}^{\mathrm{perf}}(X)$ and $\mathbf{D}^b_{\mathrm{coh}}(X)$. *Ann. Math. (2)*, 193(3):689–732, 2021.

[Nee23] Amnon Neeman. Finite approximations as a tool for studying triangulated categories. In *International congress of mathematicians 2022, ICM 2022, Helsinki, Finland, virtual, July 6–14, 2022. Volume 3. Sections 1–4*, pages 1636–1658. Berlin: European Mathematical Society (EMS), 2023.

[Nee24a] Amnon Neeman. Bounded t-structures on the category of perfect complexes. *Acta Math.*, 233(2):239–284, 2024.

[Nee24b] Amnon Neeman. Obstructions to the existence of bounded t-structures. In *Triangulated categories in representation theory and beyond. The Abel symposium 2022, Ålesund, Norway, June 6–10, 2022*, pages 195–215. Cham: Springer, 2024.

[Nee25] Amnon Neeman. Excellent metrics on triangulated categories, and the involutivity of the map taking $\mathcal{S}$ to $\mathfrak{S}(\mathcal{S})^{\mathrm{op}}$. arXiv:2505.09120, 2025.

[Nee26] Amnon Neeman. Triangulated categories with a single compact generator, and two Brown representability theorems. *Invent. math.*, available online here, 2026.

[Rou08] Raphaël Rouquier. Dimensions of triangulated categories. *J. K-Theory*, 1(2):193–256, 2008.

[Sch06] Marco Schlichting. Negative K-theory of derived categories. *Math. Z.*, 253(1):97–134, 2006.

[Sta24] The Stacks Project Authors. *Stacks Project*. `https://stacks.math.columbia.edu`, 2024.